KAIZEN MATHS

Lesson starter

Work, Energy and Power

Work-Energy Applications

Name

S1

A body of mass 8 kg has initial speed 5 m s^{-1}. 476 J of work is done on it. Find its final speed.

S2

A vehicle of mass 500 kg travelling at 12 m s^{-1} is brought to rest in 60 m by a constant braking force. Find the braking force.

S3

A constant driving force of 120 N moves an object at a steady speed of 8 m s^{-1}. Find the power developed.

S4

A lift raises a load of mass 20 kg vertically at a constant speed of 0.5 m s^{-1}. Find the power needed to lift the load. Take g = 9.8 m s^{-2}.

Show your working

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Exit ticket

Work, Energy and Power

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E1

A machine takes in 1200 J of energy and transfers 1080 J usefully. Find its efficiency.

E2

A body of mass 10 kg has initial speed 6 m s^{-1}. 945 J of work is done on it. Find its final speed.

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A vehicle of mass 1000 kg travelling at 15 m s^{-1} is brought to rest in 50 m by a constant braking force. Find the braking force.

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MEC-8E86-L2 · E

Show your working

Teacher answers & working

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S1

A body of mass 8 kg has initial speed 5 m s^{-1}. 476 J of work is done on it. Find its final speed.

Answer:

12 m s^{-1}

Working:

Work done by the resultant force equals the increase in kinetic energy.

$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$476 = \frac{1}{2} \times 8 \times v^2 - \frac{1}{2} \times 8 \times 5^2$

$476 = 4v^2 - 100$

$4v^2 = 576$

$v^2 = 144$

$v = 12$ m s^{-1}

S2

A vehicle of mass 500 kg travelling at 12 m s^{-1} is brought to rest in 60 m by a constant braking force. Find the braking force.

Answer:

600 N

Working:

Work done by the braking force equals the loss of kinetic energy.

$Fs = \frac{1}{2}mu^2$

$F \times 60 = \frac{1}{2} \times 500 \times 12^2$

$60F = 36000$

$F = \frac{36000}{60}$

$F = 600$ N

S3

A constant driving force of 120 N moves an object at a steady speed of 8 m s^{-1}. Find the power developed.

Answer:

960 W

Working:

When force and velocity are in the same direction, power = force multiplied by velocity.

$P = Fv$

$P = 120 \times 8$

$P = 960$ W

Teacher answers & working

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S4

A lift raises a load of mass 20 kg vertically at a constant speed of 0.5 m s^{-1}. Find the power needed to lift the load. Take g = 9.8 m s^{-2}.

Answer:

98 W

Working:

At constant speed, the lifting force equals the weight mg.

$F = mg$

$F = 20 \times 9.8 = 196$ N

$P = Fv$

$P = 196 \times 0.5$

$P = 98$ W

E1

A machine takes in 1200 J of energy and transfers 1080 J usefully. Find its efficiency.

Answer:

90%

Working:

Efficiency is useful output divided by total input, then multiplied by 100%.

Efficiency = $\frac{useful output}{total input} \times 100\%$

Efficiency = $\frac{1080}{1200} \times 100\%$

Efficiency = 90%

E2

A body of mass 10 kg has initial speed 6 m s^{-1}. 945 J of work is done on it. Find its final speed.

Answer:

15 m s^{-1}

Working:

Work done by the resultant force equals the increase in kinetic energy.

$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$945 = \frac{1}{2} \times 10 \times v^2 - \frac{1}{2} \times 10 \times 6^2$

$945 = 5v^2 - 180$

$5v^2 = 1125$

$v^2 = 225$

$v = 15$ m s^{-1}

Teacher answers & working

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E3

A vehicle of mass 1000 kg travelling at 15 m s^{-1} is brought to rest in 50 m by a constant braking force. Find the braking force.

Answer:

2250 N

Working:

Work done by the braking force equals the loss of kinetic energy.

$Fs = \frac{1}{2}mu^2$

$F \times 50 = \frac{1}{2} \times 1000 \times 15^2$

$50F = 112500$

$F = \frac{112500}{50}$

$F = 2250$ N